



PRODUCT & EXPERIMENTATION REPORT

GrowthOS

An Evolutionary Conversion Lab for Explainable Landing-Page Optimization

Run behavioral stress tests. Evolve a stronger page.

Product thesis

GrowthOS treats landing pages as competing growth hypotheses. It models how distinct behavioral cohorts interact with each page, evaluates pages using conversion and engagement signals, then assembles and validates a stronger next version with full lineage and rationale.

Prepared as a comprehensive product, experiment, and implementation report.

Scope: Scholé AI use case • college students preparing for exams • synthetic behavioral simulation

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OVERVIEW

1. Executive Summary

GrowthOS is a self-optimizing landing-page experimentation environment designed to compress the traditional growth loop from weeks to minutes. Instead of running a single A/B test and waiting for live traffic, it launches five structurally distinct landing-page hypotheses for the same product, simulates behavior from multiple audience cohorts, measures performance, extracts explainable findings, and produces a new evidence-backed page that is validated again against the same behavioral model.

The product under test is Scholé AI, an adaptive study tutor for college students preparing for exams. Keeping the audience and product fixed makes the core variable clear: not “who should we target,” but “what growth philosophy, message, sequence, and CTA experience should we use for this audience.”

Core product promise

GrowthOS stress-tests landing-page hypotheses against behavioral cohorts, learns which page traits earn attention and trust, then assembles a stronger next version.

Capability	What the system does	Why it matters
Hypothesis testing	Compares five pages built around different behavioral levers.	Tests strategic beliefs rather than cosmetic copy changes.
Behavioral simulation	Models 10,000 synthetic visits per page across five cohorts.	Produces directional insight before spending real acquisition traffic.
Fitness evaluation	Combines conversion with engagement, scroll depth, time, and bounce signals.	Balances business outcomes with leading indicators.
Explainable evolution	Selects, recombines, and mutates page traits using recorded evidence.	Shows why Gen 2 changed rather than presenting a black-box output.
Validation retest	Reruns simulation with Gen 2 included.	Closes the loop by testing whether the evolved page outperforms its parents.

STARTING POINT

2. Assignment Requirements and Product Framing

The assignment asked for a hosted web app that explores how landing pages can improve through experimentation and simulated user behavior. It required at least five landing pages for the same product, a simulated-data comparison, a determination of which versions performed best, new variations informed by learning, and a clear view of what changed and why.

The project was intentionally framed as a mini autonomous growth lab rather than a static landing-page gallery. That framing created a coherent product story: five initial growth beliefs enter the system, behavioral evidence applies pressure, stronger traits are selected, and an evolved page is validated.

Requirement	GrowthOS response
At least five landing pages	Five Gen 0 candidates, each with distinct copy, structure, CTA style, and behavioral hypothesis.

Requirement	GrowthOS response
Simulated user behavior	A seeded persona simulation models attention, resonance, bounce, dwell, scroll, CTA clicks, and conversion.
Compare performance	The lab reports conversion, confidence intervals, Growth Fitness Score, cohort conversion, section dwell/reach, and friction signals.
Generate variation(s)	An evolution engine creates an Evolved Candidate (Gen 2) by recombining evidence-backed sections and applying logged mutations.
Explain changes	Selection Findings, provenance, mutations, and the Page Evolution Map show parent sources and reasons for every Gen 2 trait.
Host the app	The final submission is intended to include a live hosted link, plus engineering and product-storytelling walkthroughs.

HOW THE PRODUCT EVOLVED

3. Product Development Journey and Iteration History

The build was deliberately iterative. Each major revision was triggered by a product-quality question: Does the system test real hypotheses? Do the simulated results have a defensible mechanism? Can a reviewer understand why Gen 2 changed? Does the interface make the core idea memorable?

Stage	Initial decision / issue	Iteration made	Result
1. Product thesis	A basic landing-page comparison could look like a school project.	Framed the product as a self-optimizing growth lab and an Evolutionary Conversion Lab.	The app became a system for learning, not a gallery of pages.
2. Experiment design	Five pages could be superficial copy swaps.	Defined five distinct growth hypotheses: outcome, companionship, proof, urgency, product-led discovery.	Every page now represents a competing theory of conversion.
3. Page architecture	Hardcoded pages would make evolution fake or fragile.	Represented pages as section configuration objects rendered through a shared system.	Gen 2 can be assembled programmatically and traced back to parent traits.
4. Simulation	Random metrics would not show real product reasoning.	Created persona cohorts with attention, trust, triggers, repels, bounce behavior, and CTA preferences.	Results are behavior-driven and reproducible, not arbitrary.
5. Evaluation	A single conversion percentage hides why a page wins.	Added confidence intervals, cohort breakdowns, dwell/reach, and a weighted Fitness Score.	The app can identify both the winner and the mechanisms behind it.
6. Evolution	A generated page could look like a random collage.	Added insight heuristics, recombination, copy mutations, and provenance.	Gen 2 has inspectable lineage and evidence-backed changes.
7. Storytelling	Charts alone did not create an emotional learning moment.	Added Selection Findings / "What the Engine Learned Overnight" before evolution.	The system visibly turns behavior into recommended actions.
8. Visual differentiation	Initial pages risked looking like one template with different copy.	Introduced variant-specific visual systems and signature artifacts for A–E.	The five hypotheses now feel like distinct product philosophies.
9. Product identity	The app lacked a unifying design language.	Adopted the Evolutionary Conversion Lab, Attention Profiler, and Page Evolution Map vocabulary.	The product now has a coherent thematic world.

Stage	Initial decision / issue	Iteration made	Result
10. Professional polish	Emojis and unqualified synthetic claims could feel assignment-like or misleading.	Shifted to restrained typography, visual primitives, and visible synthetic-data disclosure.	The experience is more credible and interview-ready.

AI-assisted development tools were used as implementation accelerators for scaffolding, component iteration, debugging, and verification. Product framing, growth hypotheses, simulation assumptions, tradeoffs, and final review decisions were actively directed and refined throughout the build.

WHAT A REVIEWER EXPERIENCES

4. Product Experience and User Journey

- 1. Enter the GrowthOS control layer:** The homepage introduces the thesis: traditional experimentation waits for traffic, while GrowthOS models behavioral pressure and learns earlier. It presents the five hypotheses as strategic bets rather than static mockups.
- 2. Inspect the Gen 0 candidates:** A reviewer can open each Scholé AI page and see a visibly different conversion philosophy, not just a different headline.
- 3. Run the Behavioral Stress Test:** The lab simulates traffic across all five candidates. A live playback and sampled event ticker make the simulation legible as behavior rather than an instantaneous black-box calculation.
- 4. Read the Attention Profiler and Fitness results:** The dashboard surfaces cohort conversion, section reach/dwell, conversion confidence, scroll behavior, bounce, time on page, and the aggregate Fitness Score.
- 5. Reveal the Gen 0 champion:** The highest-fit candidate is surfaced only after simulation and scoring. The result is calculated from synthetic behavior and is repeatable for the same experiment seed.
- 6. Review Selection Findings:** The system states the observed behavior, supporting evidence, and recommended action. This is the causal bridge between analytics and page changes.
- 7. Inspect the Page Evolution Map:** A visual lineage view shows how the Evolved Candidate inherits page traits from its parents and why each trait was selected.
- 8. Validate the Evolved Candidate:** Gen 2 is simulated again, compared with the Gen 0 champion, and surfaced as a validated next page only if results support that claim.

THE FIVE GEN 0 GROWTH HYPOTHESES

5. Experiment Design

The experimental unit is a landing-page configuration. The product, audience, price framing, and overall objective remain constant. The strategic variable is the conversion philosophy: the message, section sequence, CTA approach, proof mechanism, density, and interaction pattern used to persuade a college student to begin using Scholé AI.

Candidate	Growth hypothesis	Primary mechanism	Expected strongest cohort	Signature artifact
A — The Outcome	Exam-stressed students convert on concrete results, not features.	Outcome framing: grade lift, time saved, study efficiency.	The Achiever	Premium outcome hero; before/after study time; grade-progress timeline.
B — The Companion	Studying is lonely; belonging beats performance claims.	Emotional reassurance, memory, nonjudgmental tutoring.	The Social Learner	Warm chat preview and playable late-night tutor conversation.

Candidate	Growth hypothesis	Primary mechanism	Expected strongest cohort	Signature artifact
C — The Crowd	Skeptical students need evidence before promises.	Authority, peer proof, ratings, testimonials, cohort-scale evidence.	The Skeptic	Live student counter, university marquee, dense review wall.
D — The Deadline	Last-minute students respond to scarcity and peer pressure.	Countdown, social momentum, urgency, action-oriented CTA.	The Last-Minute Student	Countdown clock, joined-today counter, activity feed, scarcity state.
E — The Sandbox	Explorers convert when they experience the product before any claim.	Interactive product demonstration and adaptive workflow visibility.	The Explorer	Terminal-style workflow plus interactive subject picker.

Each candidate is represented as a sequence of typed sections: hero, social proof, features, demo, pricing, FAQ, CTA, and urgency bar. The configuration-driven architecture is the key engineering decision that makes Page Evolution Map and Gen 2 recombination real: the system can choose and reorder sections as data, rather than pretending to evolve hardcoded JSX.

Experiment scope

This is a directional, pre-live-traffic decision sandbox. It does not claim to replace real experimental evidence. The purpose is to test and prioritize hypotheses, expose tradeoffs, and produce a more deliberate candidate for live validation.

HOW SYNTHETIC USER BEHAVIOR IS GENERATED

6. Behavioral Simulation Model

GrowthOS simulates 10,000 users per candidate, for 50,000 Gen 0 visits per experiment run. Every visitor is assigned to one of five behavioral cohorts using weighted sampling. The cohort selection, attention draw, event path, and conversion outcome are generated from a seeded pseudorandom source, so the same experiment seed reproduces the same run.

Behavioral cohort	Mix	Attention budget	Trust threshold	Main triggers	Main resistance
The Skeptic	20%	45s ± 15s	0.62	Social proof, pricing clarity	Urgency bars, hype-heavy hero claims
The Achiever	25%	35s ± 10s	0.52	Outcome-oriented hero, pricing clarity	Long features, FAQ friction
The Last-Minute Student	20%	20s ± 8s	0.35	Urgency, immediate CTA	Long proof blocks, demo friction
The Explorer	15%	90s ± 25s	0.50	Interactive demo, product features	Urgency, pricing interruption
The Social Learner	20%	50s ± 15s	0.50	Social proof, FAQ reassurance	Purely technical product detail

The simulator uses several linked concepts:

- 1. Persona mix:** The estimated composition of the modeled audience. It determines how often each cohort is sampled. The mix sums to 100%, preventing any cohort from being double-counted.
- 2. Attention budget:** The number of seconds a simulated visitor is willing to spend before running out of patience. It is sampled from a Gaussian distribution and has a minimum bound so every visitor has a viable path.

- 3. Hero gate:** A fast opening-message test. A visitor can bounce early if the hero or opening experience does not resonate with their cohort. It represents the practical “three-second test” of landing-page relevance.
- 4. Resonance:** A running measure of fit between a visitor and the page. Trigger sections increase it; repellent sections decrease it. It affects both bounce risk and CTA probability.
- 5. Section dwell:** The modeled time spent on a section. It is based on section type, content length, cohort interest, and remaining attention budget.
- 6. Bounce hazard:** The probability of leaving at a given stage. It increases with baseline impatience and repellent content, and decreases when the page has earned resonance.
- 7. CTA propensity:** The probability that a visitor clicks the first meaningful conversion action they encounter. It uses a logistic response to resonance relative to the cohort’s trust threshold and accounts for CTA style.
- The session path is modeled as:

Simulation loop

Draw cohort → draw attention budget → evaluate hero gate → walk sections in order → record dwell / reach / resonance → test CTA → test bounce → record conversion, scroll depth, time, and exit point.

SIMULATION DETAIL

6.1 CTA styles and their modeled effect

CTA style	Modeled implication
Hero primary	A clear high-intent action immediately available after a concise value proposition.
Sticky bar	A persistent conversion path, particularly relevant for urgency-sensitive users.
Inline repeated	Multiple opportunities to act after sequential proof blocks.
Footer	A lower-pressure CTA that rewards deeper reading and a longer trust-building journey.
Demo embedded	The product experience itself acts as the conversion mechanism.

HOW PERFORMANCE IS MEASURED

7. Evaluation Framework and Metrics

GrowthOS deliberately avoids treating conversion as the only signal. Conversion is the primary business outcome, but engagement signals are useful for understanding why a page wins, where it loses visitors, and what should be inherited by Gen 2.

Metric	Definition	How it is used
Conversion rate	Converted visitors divided by simulated visitors. In this model, conversion means CTA click / signup intent.	Primary outcome; highest weight in Fitness Score; used to rank CTA performance.
Wilson 95% interval	A confidence interval around a binary conversion proportion.	Shows uncertainty around the estimate and avoids overconfidence from a plain percentage.

Metric	Definition	How it is used
Bounce rate	Share of visitors who exit before converting.	Detects friction and is inverted in the Fitness Score.
Average scroll depth	Average fraction of page sections reached.	Shows how far users progress through the conversion path.
Average time on page	Mean modeled session time in seconds.	Indicates sustained attention, interpreted alongside conversion.
Engagement / resonance	Average cumulative fit between cohort motivations and page sections.	Captures qualitative message-page fit in the synthetic model.
Section mean dwell	Average time among visitors who reached a section.	Used to identify attention-holding content and possible candidates for promotion.
Section reach rate	Share of all visitors who reached a section.	Used to identify sections hidden behind early friction or abandonment.
Cohort conversion	Conversion rate broken out by behavioral cohort.	Reveals segmentation effects and tradeoffs that aggregate scores can hide.

COMPOSITE RANKING

7.1 Growth Fitness Score

The Growth Fitness Score (GFS) is a 0–100 composite measure computed relative to the current set of candidates. Each raw metric is min-max normalized across candidates before weighting. This produces a useful within-run ranking, but it should not be compared blindly across entirely different experimental setups.

GFS formula

$$\text{GFS} = 100 \times [0.40 \times \text{normalized conversion} + 0.20 \times \text{normalized engagement} + 0.15 \times \text{normalized scroll depth} + 0.15 \times \text{normalized time on page} + 0.10 \times \text{normalized } (1 - \text{bounce rate})].$$

Weight	Rationale
40% Conversion	The business objective is to move a qualified visitor to the next meaningful action.
20% Engagement	Strong fit and attention are useful leading indicators when conversion alone does not explain behavior.
15% Scroll depth	A page must keep visitors moving through the intended conversion sequence.
15% Time on page	Time can indicate learning and consideration, but is intentionally lower-weighted because long time can also signal confusion.
10% Inverse bounce	Early exits matter, but bounce is a diagnostic rather than the primary outcome.

FROM EVIDENCE TO AN EVOLVED CANDIDATE

8. Explainable Evolution Engine

GrowthOS does not generate a next page by randomly remixing content. It applies a rules-based, inspectable learning process that translates results into findings, recombines the strongest traits, makes small logged mutations, and validates the result again.

THE LEARNING LAYER

8.1 Selection Findings

The engine scans results for patterns and returns structured findings with a principle, evidence, magnitude, recommended action, and source candidate. The interface presents this chain as: observed behavior → evidence → recommended action.

Heuristic	Signal detected	Potential action
Promote high-dwell sections	Section dwell correlates positively with conversion across candidates.	Move that section type earlier in the Evolved Candidate.
Cut skipped sections	A section has low reach relative to the page average while bounce is high.	Compress or remove content that delays the conversion path.
Choose CTA style	A CTA style has the highest average conversion across tested candidates.	Adopt the strongest CTA copy and style.
Address underconversion	A cohort converts poorly on every candidate.	Add or emphasize a section aligned to that cohort's strongest trigger.
Urgency sequencing	Urgency lifts Last-Minute conversion but lowers Skeptic performance.	Keep urgency, but place it after proof / trust rather than at the opening.
Demo-path optimization	A demo earns high dwell but the page converts only mid-pack.	Keep the demo but shorten the path from demonstration to CTA.

HOW GEN 2 IS BUILT

8.2 Recombination, mutation, and lineage

- 1. Hero selection:** The system selects the hero content from the page with the best hero-gate pass rate weighted by downstream conversion.
- 2. Section selection:** The strongest social proof, feature, and pricing sections are selected using observed dwell, reach, and page conversion context.
- 3. Trust-aware sequencing:** Social proof leads the middle of Gen 2 when evidence supports trust building, while urgency can be retained below that proof block rather than leading with it.
- 4. Content compression:** Oversized lists can be trimmed to preserve a shorter path to the CTA when behavior indicates friction or low utility.
- 5. Copy mutation:** Small changes are logged, such as adding a concrete number, shortening an overlong subheadline, or adding an urgency cue only when the data supports it.
- 6. Provenance:** Each Gen 2 section carries a parent source and a reason, so the page remains inspectable. This is Page Lineage, not opaque generation.

Page Evolution Map

The signature visual represents Gen 0 candidates as ordered page blueprints and the Evolved Candidate as an evidence-backed assembly. It makes inheritance visible: which page trait moved forward, where it came from, and why it was selected.

CLOSING THE LOOP

8.3 Validation Retest

After Gen 2 is generated, the system reruns the simulation with a fresh deterministic stream. The dashboard compares the Gen 0 champion with the Evolved Candidate using actual calculated Fitness Scores. The UI should not claim a lift before this retest has occurred.

WHY THE INTERFACE LOOKS THE WAY IT DOES

9. Visual Identity and Product Theme

The final visual direction is “GrowthOS: The Evolutionary Conversion Lab.” This theme makes the product’s mechanics legible without overstating the implementation as a literal biological genetic algorithm. The system is evolution-inspired: behavioral pressure selects useful traits, traits are recombined, small mutations are logged, and the result is retested.

Product language	Meaning in GrowthOS
Gen 0 Candidates / Growth Hypotheses	The five initial landing-page configurations.
Behavioral Cohorts	The persona segments that supply distinct motivations and selection pressure.
Behavioral Stress Test	The simulated traffic run across candidates.
Fitness Score	The weighted composite used to rank page performance.
Attention Profiler	The heatmap / dwell / reach view showing where cohorts engage, lose momentum, or exit.
Selection Findings	Evidence-backed insights that turn behavior into concrete recommendations.
Page Evolution Map	The lineage visualization showing how Gen 2 inherits selected page traits.
Evolved Candidate	Gen 2, assembled from evidence and subject to validation.
Validation Retest	The post-evolution simulation comparing Gen 2 with Gen 0.

AVOIDING A “SAME TEMPLATE, DIFFERENT COPY” PROBLEM

9.1 Distinct visual philosophies for the five candidates

Candidate	Visual system	Why it reinforces the hypothesis
A — Outcome	Dark premium outcome surface, oversized grade lift, before/after study-time comparison, score progression timeline.	Focuses attention on measurable transformation, efficiency, and aspirational results.
B — Companion	Warm, low-pressure palette, conversational chat bubbles, a replayable 2 AM support interaction.	Makes emotional support tangible and demonstrates persistent tutoring memory.
C — Crowd	Live enrollment count, university marquee, ratings, high testimonial density, review metadata.	Floods the page with authority and peer validation for skeptical visitors.
D — Deadline	High-contrast countdown clock, sticky urgency treatment, live activity feed, capacity language.	Creates immediacy and social momentum for a deadline-sensitive cohort.
E — Sandbox	Terminal-style analysis flow and interactive subject selection with adaptive response state.	Lets product curiosity do the persuasion before a broader product claim.

The brand system uses professional typography, restrained accent colors, dots, arrows, labels, borders, and subtle motion. Decorative emojis are avoided in visible product UI; visual states such as selection, live activity, and validation are communicated through hierarchy and interface primitives instead.

IMPLEMENTATION APPROACH

10. Technical Architecture

GrowthOS is a Next.js App Router application using TypeScript and Tailwind CSS. The interaction layer uses Framer Motion for restrained animation, Recharts for dashboards, and canvas-confetti for the two outcome moments: the Gen 0 winner reveal and confirmed Gen 2 validation. The product is intentionally database-free for the take-home: experiment state is deterministic and in-memory, while the generated Gen 2 configuration is persisted locally for page preview.

Layer	Responsibilities	Key implementation choice
Configuration layer	Defines candidate pages, section order, copy, themes, CTA styles, and hypotheses.	Landing pages are data objects, not isolated hardcoded pages.
Renderer layer	Renders generic or variant-specific hero, proof, demo, feature, pricing, FAQ, CTA, and urgency components.	Supports rich candidate-specific experiences while preserving common data contracts.
Simulation layer	Generates seeded users, cohort paths, dwell, resonance, bounce, CTA events, and conversion.	Independent deterministic random streams per variant enable reproducible comparisons.
Metrics layer	Aggregates behavior, computes Wilson intervals, cohort matrix, section metrics, and Fitness Score.	Supports both headline ranking and diagnostic analysis.
Evolution layer	Extracts findings, chooses parent sections, applies mutations, attaches provenance, builds Gen 2.	Explainable rules and lineage make the output defensible.

Layer	Responsibilities	Key implementation choice
Experience layer	Runs the lab pipeline, live simulation playback, charts, Selection Findings, Page Evolution Map, and validation.	Turns algorithmic behavior into an understandable product narrative.

FROM HYPOTHESIS TO VALIDATION

10.1 Data flow

End-to-end flow

Variant configuration → seeded simulation result → per-variant metrics → Fitness Scores + cohort/section analysis → Selection Findings → Gen 2 configuration + provenance → validation retest → comparison artifact.

HOW THE PRODUCT IS CHECKED

11. Testing, Reliability, and Demo Readiness

Testing serves two purposes: technical correctness and storytelling integrity. A Growth Engineer submission is not only about whether a page compiles; it is also about whether the experimental claims shown in the demo match what the system actually calculates.

Test category	Checks
Static quality	Run npm lint to catch TypeScript / React / style issues, then fix all errors before delivery.
Production readiness	Run npm build to catch route, server, hydration, and production compilation issues.
Route smoke test	Verify homepage, lab, variants A–E, and Gen 2 route after evolution.
Experiment flow	Run a full path: candidate display → simulation → metrics → winner → Selection Findings → Page Evolution Map → evolution → validation.
Seed behavior	Rerun with the same seed to verify reproduction; reroll seed to ensure rankings are not a hidden hardcoded value.
Variant differentiation	Confirm each candidate renders its signature visual system and interaction, not only generic sections.
Responsive QA	Check the lab, cards, map, and individual variants at desktop and mobile dimensions.
Integrity check	Confirm synthetic-data disclosure is visible and that no fictional school/user/grade numbers are presented as real outcomes.
Hosted verification	Open the live app in an incognito window and retest the full flow before submission.

WHAT THE SYSTEM DOES NOT CLAIM

12. Experiment Integrity, Assumptions, and Limitations

The strongest way to present GrowthOS is with clear boundaries. It is a high-velocity hypothesis and prioritization tool, not a substitute for actual customer experimentation.

Assumption / limitation	Why it matters	How to address it with real traffic
Synthetic cohorts	Personas are designed assumptions, not observed populations.	Calibrate parameters from surveys, event data, session replays, CRM segments, and cohort analysis.
Rule-based resonance	The model encodes a directional belief about what users respond to.	Learn weights from real experiment results or fit a predictive model with held-out validation.
CTA click as conversion	The simulation models intent, not downstream payment, activation, or retention.	Connect real funnel events: click → signup → activation → retained user.
Fitness Score is relative	Min-max normalization makes the score useful within a run but not universal.	Track raw metrics alongside scores and compare equivalent experiment cohorts.
Synthetic claims	Testimonials, ratings, schools, counts, and grade lifts are illustrative.	Use approved real proof or replace with clearly labeled simulated content.
No persistent experiment store	The take-home favors determinism and simplicity over production history.	Add a database for experiments, configurations, cohorts, runs, findings, and approvals.
No causal live allocation	The system ranks synthetic candidates but does not yet dynamically allocate traffic.	Use controlled live tests or a contextual bandit / Thompson-sampling policy with guardrails.

Disclosure language

Synthetic behavioral experiment data for demonstration — not real customer outcomes.

HOW GROWTHOS COULD EVOLVE BEYOND

13. Production Roadmap

- 1. Instrument real behavior:** Connect product analytics (for example, first-party events or an analytics platform) so section impressions, scroll, CTA events, and conversion outcomes feed the existing metrics schema.
- 2. Persist experiment history:** Store candidate configurations, seeds, runs, findings, approvals, and version lineage so teams can track learning over time.
- 3. Calibrate cohort parameters:** Use real segment data to update attention budgets, trust needs, and interaction propensity. Keep a human-review process for assumptions.
- 4. Add live validation governance:** Use controlled rollouts, holdouts, confidence thresholds, statistical guardrails, and rollback criteria before automatically promoting a page.
- 5. Use adaptive traffic allocation:** Once live metrics are available, introduce multi-armed bandit allocation for promising variants while preserving exploration and minimum-sample protections.
- 6. Extend trait evolution:** Allow an approved content-generation layer to propose a limited number of new copy or layout mutations, then subject each one to the same explainable validation process.

COMMON TERMINOLOGY

Appendix A. Glossary

Term	Meaning
Candidate / Variant	One landing-page configuration being compared against others.
Gen 0	The initial population of five page hypotheses.
Evolved Candidate / Gen 2	The page assembled from evidence-backed parent traits and small logged mutations.
Behavioral cohort	A modeled audience segment with distinct motivations and interaction tendencies.
Hero gate	An early resonance / bounce check representing the opening-message test.
Attention budget	The simulated amount of time a visitor is willing to spend before leaving.
Resonance	The running fit score between a page's content and a visitor's motivations.
Dwell	Modeled time spent in a given section by visitors who reach it.
Reach rate	Percentage of all visitors who reach a particular section.
Bounce hazard	The conditional probability of exiting a page at a given stage.
CTA propensity	The modeled likelihood of clicking a conversion action when it is encountered.
Wilson interval	A confidence interval for a binary rate such as conversion.
Fitness Score	A weighted, relative ranking combining conversion and leading indicators.
Selection Finding	An explainable insight that maps observed behavior to a recommended action.
Page Lineage	The record of which parent candidate contributed a Gen 2 section and why.
Validation Retest	A follow-up simulation that tests the Evolved Candidate against its Gen 0 parents.

IMPLEMENTATION REFERENCES

Appendix B. Source Files and Ownership Map

File / area	Role in the system
lib/variants.ts	Defines section types, candidate themes, CTA styles, hypotheses, copy, and initial variant configuration.
lib/personas.ts	Defines behavioral cohorts, population mix, attention, trust, triggers, repels, bounce baseline, and urgency sensitivity.
lib/simulation.ts	Runs seeded behavioral simulations and records visitor-level events.
lib/metrics.ts	Computes confidence intervals, section aggregates, cohort conversion, and Fitness Scores.
lib/evolution.ts	Extracts Selection Findings, recombines sections, applies mutations, and records lineage for Gen 2.
components/landing/*	Renders generic and candidate-specific landing-page sections.
components/lab/*	Presents the pipeline, playback, metrics, Attention Profiler, findings, Page Evolution Map, and validation.
app/page.tsx	GrowthOS product homepage and product narrative entry point.
app/lab/page.tsx	Primary laboratory / experiment orchestration experience.
app/variant/[id]/page.tsx	Renders Gen 0 candidate pages and the locally persisted Gen 2 preview.

DELIVERY READINESS

Appendix C. Final Submission Checklist

Checklist item	Completion criterion
Hosted app	Public URL loads in an incognito window and supports the full lab flow.
Build quality	pnpm lint and pnpm build both pass cleanly.
Visual QA	Variants A–E render with their unique signature experiences; Gen 2 renders after evolution.
Experiment integrity	Synthetic-data disclosure is visible; no illustrative numbers are implied to be real customer outcomes.
Engineering video	Explains assumptions, config-driven architecture, simulation, metrics, evolution, and limitations.
Pitch video	Shows five hypotheses, behavioral pressure, Selection Findings, Page Evolution Map, and Validation Retest.
Submission message	Includes hosted app link, video links, and optional repository / README link.